Task Description: Samsung Phone Query and Review System

You are part of a development team at a tech company tasked with creating an intelligent system that allows users to easily gather detailed information, ask questions, and read reviews about Samsung smartphones. The system will leverage web scraping, chatbot interaction, multi-agent systems, and API integration. Here's what needs to be done:

1. Scrape Samsung Phone Data:

- **Objective**: Collect detailed specifications and features for 10-15 Samsung phones (e.g., Galaxy S21, S22, S23, etc.) from **GSMArena** by web scraping. (BeautifulSoup, Scrapy, Selenium)
- **Details**: Your scraper should capture key phone specifications like display size, camera specs, battery life, processor, and other features.
- **Database**: Store the scraped data in a structured database (e.g., **PostgreSQL** or **MySQL**) to make it easily accessible for later retrieval.

2. Build a Conversational Chatbot Using LLM:

- **Objective**: Develop a chatbot (use open source models, RAG) that can answer a wide range of questions about Samsung phones, including their specifications, features, pricing, and comparisons.
- **Sample Queries**:
    - "What are the camera specs of the Samsung Galaxy S23?"
    - "Which Samsung phone has the best battery life?"
    - "How does the Galaxy S23 compare to the S22 in terms of performance?"

3. Implement a Multi-Agent System:

- **Objective**: Develop a multi-agent system (CrewAi, AutoGen, LangChain) where different agents specialize in fetching specific types of data and generating reviews.
    - One agent retrieves detailed phone specifications from the database.
    - An agent generates a detailed product review by combining the specifications.

4. API Integration for Query Handling:

- **Objective**: Create an API (FastAPI, Flask) that allows users to interact with the system, asking questions about Samsung phones and receiving accurate responses.

Execution Plan

1. **Step 1: Scrape Data and Store It**
    - Begin by scraping data from **GSMArena** using **BeautifulSoup** or **Scrapy**. Focus on 10-15 popular Samsung models.
    - Create a **PostgreSQL** database to store the scraped data in an organized manner (i.e., tables for phones, specifications, prices, etc.).
2. **Step 2: Build the Chatbot**
    - Integrate **OpenAI** or **Open source models ([Hugging Face])** for the chatbot. The chatbot should be able to answer questions like "What is the screen size of the Galaxy S22?"
    - Connect the chatbot to your database, enabling it to dynamically fetch phone specifications using **RAG** based on user input.
3. **Step 3: Implement Multi-Agent System**
    - Build a multi-agent system where one agent retrieves the technical data, another generates comprehensive reviews.
4. **Step 4: Develop the API**
    - Create a **FastAPI** or **Flask** API with endpoints that accept user queries in and return responses.

Assessment Criteria

- **Data Scraping**: Accuracy, completeness, and structure of the scraped data stored in the database.
- **Chatbot**: Effective handling of diverse user queries, with relevant and accurate responses.
- **Multi-Agent System**: The effectiveness of agents working together to provide detailed and dynamic product reviews.
- **API Integration**: Seamless interaction with the API, handling multiple queries and providing accurate responses.

Submission Guidelines:

1. **GitHub**: Create a GitHub repository and upload your project files.
2. **Google Drive Video**: Record a 5-10 minute video explaining the project, then upload it to Google Drive and get a shareable link.
3. **Email**: Send an email to **hr02@gtrbd.com** **&** **hr01@gtrbd.com** with the links to your GitHub repository and the Google Drive video.